



nology development in the LED space is so fast that an LED lighting product that is new today could be outdated the next month. So it is very difficult to say what's new in the market today.

A majority of commercial lighting involves recessed or canned lighting, which releases very little heat. These provide a focused, high-intensity light where it is needed the most. They can be operated by dimmer switches that help tailor the lighting to the needs of the individual.

AMKA Lighting has added new products to its wide range of LED recessed downlights, spotlights, surface lights and power LED lamps, which the company claims to be a replacement of the MR16 halogen lamp.

"These power LED lamps can save at least 85 per cent of the energy normally used and have a working life of 20,000 hours," informs Sanjive Jain, managing director, and S.C. Joneja, director, AMKA Lighting.

Binay Opto's model 600³ power LED luminaire is designed for recessed fitting and is suitable for installation into a panel grid. "It offers 95 per cent luminaire efficiency and saves 75 per cent more power than comparable fluorescent tubelights," explains Kumar K. Rohatgi, founder and technical

advisor, Binay Group.

LED retrofit bulbs, which can conveniently replace the GLS bulb and be installed on existing lighting fixtures without any additional expense, are new in the market. These globe-shaped bulbs also come in RGB colour changing models and with a remote control.

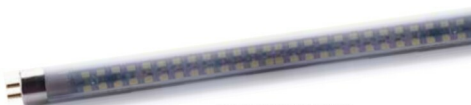
In October 2010, Philips Lighting India launched a range of LED bulbs for consumers in the Indian market. These indigenously developed LED bulbs adhere to India-specific environmental and infrastructural challenges such as voltage fluctuations, energy spikes and an irregular power supply.

"Unlike other lighting technologies that are electrical, LED lights are electronic devices. LED bulbs, like other LED sources, are one of the most sustainable lighting solutions. These are based on low-voltage technology and have a lifespan of 50,000 hours," says Aditya Khosla, director, Sanarti.

New additions from Sanarti include LED T5 and T8 tubelight series, LED bulb series and LED panel light series. T5 and T8 tubelights are an energy-efficient replacement for traditional fluorescent tubes and have an option of inbuilt or separate ballast.

Almost every LED player offers different kinds of downlights. "We have developed downlights using a special diffuser where the light source is invisible. Another new product is the MR16 version which is dimmable," informs G. Gururaja, director-operations, Avni Energy Solutions.

"An MR16 LED bulb is a multifaceted reflector (MR) that is usually high-powered but consumes only 4 watts. It emits the same light output as a standard 50W bulb," informs Jawed Khan, CEO, Hovel LED Lighting. The company has a series of MR16-compatible modules.



MIRC Electronics' LED tubelight

Binay Opto's retrofit power LED tubelights can replace an existing T8 fluorescent tube. "We have filed a patent for this product. There is no need to remove the electromagnetic choke. The unit can run through the choke or directly without the choke," explains Rohatgi.

Other players who offer tubelights are Sanarti, AMKA and MIRC Electronics.

"Our LED lights are anti-glare with a higher light output and lesser heat dissipation," says Avnish Jauhari, business head, lighting, MIRC Electronics.

High bay LED lights are also in demand, as these lights have a much longer working life than conventional fluorescent and incandescent lights. On an average, you can expect these lights to last ten times as long as a typical compact fluorescent bulb. They are usually used in factories and warehouses, as industrial shed lights, front lights, hoarding lights, etc.

"High bay LED lights have found wide acceptability in commercial applications replacing traditional CFL/MH high bay lights, which consume more power and generate a lot of heat," says Kapil Sood, managing director, Amptek LED Lighting.

"Our bay lights have 100 per cent application efficiency and 80 per cent power efficiency. Their lifespan is 50,000 hours," says Shyam Jindal, managing director, Olive Exports.

Amptek LED Lighting, Sujana and



Sanarti's LED bulb